

Antonio Conopio

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EDUCATION

Bachelor of Computing, Software Engineering (Co-op), Minor in Math

Sep 2023 – Apr 2028

University of Guelph

Guelph, ON

- **Cumulative GPA:** 3.89 | Dean's List: 2023 - present

EXPERIENCE

Data Integration Intern (API Developer)

Sep 2025 – Aug 2026

Co-operators

Guelph, ON

- Migrated the team's internal API template from R/Plumber to Python/FastAPI, reducing average response latency by 33%, improving modularity and input validation, and raising base test coverage from 40% to 97%.
- Owned API feature development from stakeholder requirements through implementation, testing, and debugging, working across technical and business teams to deliver production-ready changes.
- Extended auto and home Risk Assist APIs with new fields, validation logic, and business rules to onboard two new business channels, increasing API call volume by 30%.
- Led testing for the National Auto Risk Assist API, one of the team's flagship projects of the year, debugging across multiple models and data transformations to validate 100% parity with the modelers' pipeline.
- Built and deployed an R/Plumber API on Posit, with an automation script that migrates 1M+ API logs to Azure Data Lake Storage (ADLS), replacing manual extraction as part of the team's cloud modernization initiative.

PROJECTS

Orion | *Go, Python, React, TypeScript, Redux, PostgreSQL, Redis, WebSockets, Docker, Kubernetes*

- Designed and built a distributed workflow orchestration platform end-to-end, supporting DAG-based job execution, real-time status updates, retries, and horizontally scalable workers.
- Engineered a Redis Streams consumer-group architecture for distributed Python workers, providing at-least-once task delivery and automatic retry handling for failed jobs.
- Built a real-time WebSocket hub in Go to push live task-status updates to the React frontend, eliminating the need for polling and improving dashboard responsiveness.
- Containerized the platform with Docker and Kubernetes, enabling horizontal worker scaling based on Redis queue depth.

Lantern | *Python, FastAPI, LangChain, LangGraph, ChromaDB, Pydantic, pytest, Docker*

- RAG service that answers natural-language questions over uploaded documents with page-level citations, declining questions the documents can't answer.
- Implemented an ingestion pipeline that chunks and embeds PDFs into ChromaDB, then extended it into a LangGraph agent with conditional retrieval, query rewriting, and multi-turn session memory.
- Built a 28-case eval suite scoring retrieval and answers separately; used it to diagnose dropped output fields and raise answer accuracy from 86% to 100% with 92% citation precision.
- Injected models through FastAPI dependencies so 61 pytest tests run against fake LLMs and an in-memory vector store, with no API keys.

HomeHero | *Hackville 2026 | SwiftUI, Spring Boot, Supabase, TCA*

- Full-stack household management app for shared tasks, shopping lists, expenses, and home data, built in 24 hours.
- Developed the SwiftUI frontend using The Composable Architecture and async/await to manage application state and asynchronous workflows.
- Implemented a Spring Boot REST API for authentication, task operations, and multi-user data synchronization.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript/JavaScript, Go, C, SQL, Swift

Frameworks: FastAPI, Spring Boot, React, Node.js, SwiftUI

Databases: PostgreSQL, Redis, Supabase, Firebase

AI & LLM: LangChain, LangGraph, ChromaDB, RAG

Cloud & DevOps: Git, Docker, Kubernetes, Azure, CI/CD

Libraries & Tools: pytest, pandas, scikit-learn